

Measuring What Counts: The Case of Rumour Stance Classification

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Abstract

Stance classification can be a powerful tool for understanding whether and which users believe in online rumours. The task aims to automatically predict the stance of replies towards a given rumour, namely *support*, *deny*, *question*, or *comment*. Numerous methods have been proposed and their performance compared in the RumourEval shared tasks in 2017 and 2019. Results demonstrated that this is a challenging problem since naturally occurring rumour stance data is highly imbalanced. This paper specifically questions the evaluation metrics used in these shared tasks. We re-evaluate the systems submitted to the two RumourEval tasks and show that the two widely adopted metrics – accuracy and macro- $F1$ – are not robust for the four-class imbalanced task of rumour stance classification, as they wrongly favour systems with highly skewed accuracy towards the majority class. To overcome this problem, we propose new evaluation metrics for rumour stance detection. These are not only robust to imbalanced data but also score higher systems that are capable of recognising the two most informative minority classes (*support* and *deny*).

1 Introduction

The automatic analysis of online rumours has emerged as an important and challenging Natural Language Processing (NLP) task. Rumours in social media can be defined as claims that cannot be verified as true or false at the time of posting (Zubiaga et al., 2018). Prior research (Mendoza et al., 2010; Kumar and Carley, 2019) has shown that the stances of user replies are often a useful predictor of a rumour’s likely veracity, specially in the case of false rumours that tend to receive a higher number of replies denying them (Zubiaga et al., 2016). However, their automatic classification is far from trivial as demonstrated by the

results of two shared tasks – RumourEval 2017 and 2019 (Derczynski et al., 2017; Gorrell et al., 2019). More specifically, sub-task A models rumour stance classification (RSC) as a four-class problem, where replies can:

- **support**/agree with the rumour;
- **deny** the veracity of the rumour;
- **query**/ask for additional evidence;
- **comment** without clear contribution to assessing the veracity of the rumour.

Figure 1 shows an example of a reply *denying* a post on Twitter.



Figure 1: Example of a *deny* stance.

In RumourEval 2017 the training data contains 297 rumourous threads about eight events. The test set has 28 threads, with 20 threads about the same events as the training data and eight threads about unseen events. In 2019, the 2017 training data is augmented with 40 Reddit threads. The new 2019 test set has 56 threads about natural disasters from Twitter and a set of Reddit data (25 threads). These datasets for RSC are highly imbalanced: the *comment* class is considerably larger than the other classes. Table 1 shows the distribution of stances per class in both 2017 and 2019 datasets, where 66% and 72% of the data (respectively) correspond to *comments*. *Comments* arguably are the

	2017	2019
support	1,004 (18%)	1,184 (14%)
deny	415 (7%)	606 (7%)
query	464 (8%)	608 (7%)
comment	3,685 (66%)	6,176 (72%)
total	5,568	8,574

Table 1: Distribution of stances per class – with percentages between parenthesis.

least useful when it comes to assessing overall rumour veracity, unlike *support* and *deny* which have been shown to help with rumour verification (Mendoza et al., 2010). Therefore, RSC is not only an imbalanced, multi-class problem, but it also has classes with different importance. This is different from standard stance classification tasks (e.g. SemEval 2016 task 6 (Mohammad et al., 2016)), where classes have arguably the same importance. It also differs from the veracity task (RumourEval sub-task B), where the problem is binary and it is not as an imbalanced problem as RSC.¹

RumourEval 2017 evaluated systems based on accuracy (ACC), which is not sufficiently robust on imbalanced datasets (Huang and Ling, 2005). This prompted the adoption of macro- $F1$ in the 2019 evaluation. Kumar and Carley (2019) also argue that macro- $F1$ is a more reliable evaluation metric for RSC. Previous work on RSC also adopted these metrics (Li et al., 2019b; Kochkina et al., 2018; Dungs et al., 2018).

This paper re-evaluates the sub-task A results of RumourEval 2017 and 2019.² It analyses the performance of the participating systems according to different evaluation metrics and shows that even macro- $F1$, that is robust for evaluating binary classification on imbalanced datasets, fails to reliably evaluate the performance on RSC. This is particularly critical in RumourEval where not only is data imbalanced, but also two minority classes (*deny* and *support*) are the most important to classify well. Based on prior research on imbalanced datasets in areas other than NLP (e.g. Yijing et al. (2016) and Elrahman and Abraham (2013)), we propose four alternative metrics for evaluating RSC. These metrics change the systems ranking for RSC in RumourEval 2017 and 2019, rewarding systems with high performance on the minority classes.

¹Other NLP tasks, like sentiment analysis are also not comparable, since these tasks are either binary classification (which is then solved by using macro- $F1$) or do not have a clear priority over classes.

²We thank the organisers for making the data available.

2 Evaluation metrics for classification

We define TP = true positives, TN = true negatives, FP = false positives and FN = false negatives, where TP_c (FP_c) is equivalent to the true (false) positives and TN_c (FN_c) is equivalent to the true (false) negatives for a given class c .

Accuracy (ACC) is the ratio between the number of correct predictions and the total number of predictions (N): $ACC = \frac{\sum_{c=1}^C TP_c}{N}$, where C is the number of classes. ACC only considers the values that were classified correctly, disregarding the mistakes. This is inadequate for imbalanced problems like RSC where, as shown in Table 1, most of the data is classified as *comments*. As shown in Section 3, most systems will fail to classify the *deny* class and still achieve high scores in terms of ACC . In fact, the best system for 2017 according to ACC (Turing) fails to classify all *denies*.

Precision (P_c) and Recall (R_c) P_c is the ratio between the number of correctly predicted instances and all the predicted values for c : $P_c = \frac{TP_c}{TP_c + FP_c}$. R_c is the ratio between correctly predicted instances and the number of instances that actually belongs to the class c : $R_c = \frac{TP_c}{TP_c + FN_c}$.

macro- $F\beta$ $F\beta_c$ score is defined as the harmonic mean of precision and recall, where the per-class score can be defined as: $F\beta_c = (1 + \beta^2) \frac{P_c \cdot R_c}{\beta^2 P_c + R_c}$. If $\beta = 1$, $F\beta$ is the $F1$ score. If $\beta > 1$, R is given a higher weight and if $\beta < 1$, P is given a higher weight. The macro- $F\beta$ is the arithmetic mean between the $F\beta$ scores for each class: $\text{macro-}F\beta_c = \frac{\sum_{c=1}^C F\beta_c}{C}$. Although macro- $F1$ is expected to perform better than ACC for imbalanced binary problems, its benefits in the scenario of multi-class classification are not clear. Specifically, as it relies on the arithmetic mean over the classes, it may hide the poor performance of a model in one of the classes if it performs well on the majority class (i.e. *comments* in this case). For instance, as shown in Table 2, according to macro- $F1$ the best performing system would be *ECNU*, which still fails to classify correctly almost all *deny* instances.

Geometric mean Metrics like the geometric mean of R :

$$GMR = \sqrt[C]{\prod_{c=1}^C R_c}.$$

	<i>ACC</i>	macro- <i>F1</i>	<i>GMR</i>	<i>wAUC</i>	<i>wF1</i>	<i>wF2</i>
Turinga	0.784 (1)	0.434 (5)	0.000 (8)	0.583 (7)	0.274 (6)	0.230 (7)
UWaterloo (Bahuleyan and Vechtomova, 2017)	0.780 (2)	0.455 (2)	0.237 (5)	0.595 (5)	0.300 (2)	0.255 (6)
ECNU (Wang et al., 2017)	0.778 (3)	0.467 (1)	0.214 (7)	0.599 (4)	0.289 (4)	0.263 (4)
Mama Edha (García Lozano et al., 2017)	0.749 (4)	0.453 (3)	0.220 (6)	0.607 (1)	0.299 (3)	0.283 (3)
NileTMRG (Enayet and El-Beltagy, 2017)	0.709 (5)	0.452 (4)	0.363 (1)	0.606 (2)	0.306 (1)	0.296 (1)
IKM (Chen et al., 2017)	0.701 (6)	0.408 (7)	0.272 (4)	0.570 (8)	0.241 (7)	0.226 (8)
IITP (Singh et al., 2017)	0.641 (7)	0.403 (8)	0.345 (2)	0.602 (3)	0.276 (5)	0.294 (2)
DFKI DKT (Srivastava et al., 2017)	0.635 (8)	0.409 (6)	0.316 (3)	0.589 (6)	0.234 (8)	0.256 (5)
majority class	0.742	0.213	0.000	0.500	0.043	0.047
all denies	0.068	0.032	0.000	0.500	0.051	0.107
all support	0.090	0.041	0.000	0.500	0.066	0.132

Table 2: Evaluation of RumourEval 2017 submissions. Values between parenthesis are the ranking of the system according to the metric. The official evaluation metric column (*ACC*) is highlighted in bold.

	<i>ACC</i>	macro- <i>F1</i>	<i>GMR</i>	<i>wAUC</i>	<i>wF1</i>	<i>wF2</i>
BLCU NLP (Yang et al., 2019)	0.841 (2)	0.619 (1)	0.571 (2)	0.722 (2)	0.520 (1)	0.500 (2)
BUT-FIT (Fajcik et al., 2019)	0.852 (1)	0.607 (2)	0.519 (3)	0.689 (3)	0.492 (3)	0.441 (3)
eventAI (Li et al., 2019a)	0.735 (11)	0.578 (3)	0.726 (1)	0.807 (1)	0.502 (2)	0.602 (1)
UPV (Ghanem et al., 2019)	0.832 (4)	0.490 (4)	0.333 (5)	0.614 (5)	0.340 (4)	0.292 (5)
GWU (Hamidian and Diab, 2019)	0.797 (9)	0.435 (5)	0.000 (7)	0.604 (6)	0.284 (5)	0.265 (6)
SINAI-DL (García-Cumbreras et al., 2019)	0.830 (5)	0.430 (6)	0.000 (8)	0.577 (7)	0.255 (7)	0.215 (7)
wshuyi	0.538 (13)	0.370 (7)	0.467 (4)	0.627 (4)	0.261 (6)	0.325 (4)
Columbia (Liu et al., 2019)	0.789 (10)	0.363 (8)	0.000 (9)	0.562 (10)	0.221 (10)	0.191 (9)
jurebb	0.806 (8)	0.354 (9)	0.122 (6)	0.567 (9)	0.229 (8)	0.120 (12)
mukundyr	0.837 (3)	0.340 (10)	0.000 (10)	0.570 (8)	0.224 (9)	0.198 (8)
nxl	0.828 (7)	0.327 (11)	0.000 (11)	0.557 (11)	0.206 (11)	0.173 (10)
WeST (Baris et al., 2019)	0.829 (6)	0.321 (12)	0.000 (12)	0.551 (12)	0.197 (12)	0.161 (11)
Xinthl	0.725 (12)	0.230 (13)	0.000 (13)	0.493 (13)	0.072 (13)	0.071 (13)
majority class	0.808	0.223	0.000	0.500	0.045	0.048
all denies	0.055	0.026	0.000	0.500	0.042	0.091
all support	0.086	0.040	0.000	0.500	0.063	0.128

Table 3: Evaluation of RumourEval 2019 submissions. Values between parenthesis are the ranking of the system according to the metric. The official evaluation metric column (macro-*F1*) is highlighted in bold.

are proposed for evaluating specific types of errors. As *FNs* may be more relevant than *FPS* for imbalanced data, assessing models using *R* is an option to measure this specific type of error. Moreover, applying *GMR* for each class severely penalises a model that achieves a low score for a given class.

Area under the ROC curve Receiver operating characteristic (*ROC*) (Fawcett, 2006) assesses the performance of classifiers considering the relation between R_c and the false positive rate, defined as (per class): $FPR_c = \frac{FP_c}{TN_c + FP_c}$. Since RSC consists of discrete classifications, *ROC* charts for each c contain only one point regarding the coordinate (FPR_c, R_c) . Area under the *ROC* curve (*AUC*) measures the area of the curve produced by the points in an *ROC* space. In the discrete case, it measures the area of the polygon drawn by the segments connecting the vertices $((0, 0), (FPR_c, R_c), (1, 1), (0, 1))$. High *AUC* scores are achieved when *R* (probability of detection) is maximised, while *FPR* (probability of false alarm) is minimised. We experiment with a weighted variation of *AUC*:

$$wAUC = \sum_{c=1}^C w_c \cdot AUC_c.$$

Weighted macro- $F\beta$ a variation of macro- $F\beta$, where each class also receives different weights, is also considered:

$$wF\beta = \sum_{c=1}^C w_c \cdot F\beta_c,$$

We use $\beta = 1$ (*P* and *R* have the same importance) and $\beta = 2$ (*R* is more important). Arguably, misclassifying *denies* and *supports* (FN_D and FN_S , respectively) is equivalent to ignore relevant information for debunking a rumour. Since *FNs* negatively impact *R*, we hypothesise that $\beta = 2$ is more robust for the RSC case.

wAUC and *wF β* are inspired by empirical evidence that different classes have different importance for RSC.³ Weights should be manually defined, since they cannot be automatically learnt.

³Similarly, previous work proposes metrics (Elkan, 2001) and learning algorithms (Chawla et al., 2008) based on class-specific mis-classification costs.

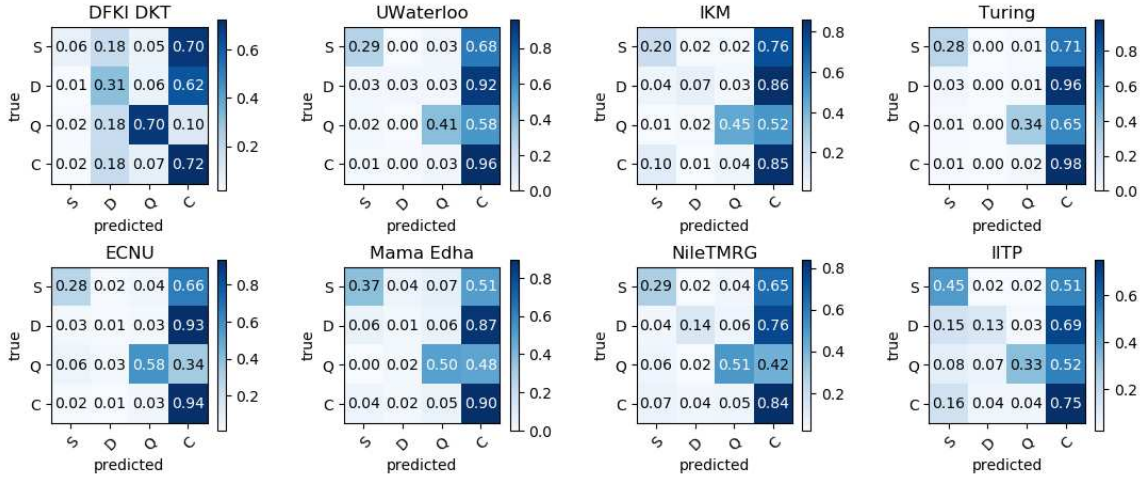


Figure 2: Confusion matrix for systems from RumourEval 2017.

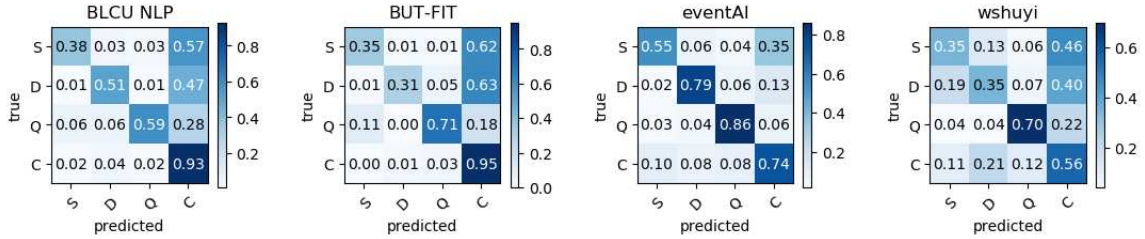


Figure 3: Confusion matrix for selected systems from RumourEval 2019. All other systems failed to classify correctly either all or the vast majority of *deny* instances.

We follow the hypothesis that *support* and *deny* classes are more informative than others.⁴

3 Re-evaluating RumourEval task A

Tables 2 and 3 report the different evaluation scores per metric for each of the RumourEval 2017 and 2019 systems.⁵ *ACC* and macro-*F1* are reported in the second and third columns respectively, followed by a column for each of the four proposed metrics. Besides evaluating the participating systems, we also computed scores for three baselines: majority class (all stances are considered *comments*), all denies and all support (all replies are classed as *deny/support*).

Our results show that the choice of evaluation metric has a significant impact on system ranking. In RumourEval 2017, the winning system based on *ACC* was Turing. However, Figure 2 shows that this system classified all *denies* in-

correctly, favouring the majority class (*comment*). When looking at the macro-*F1* score, Turing is classified as fifth, whilst the winner is ECNU, followed by UWaterloo. Both systems also perform very poorly on *denies*, classifying only 1% and 3% of them correctly. On the other hand, the four proposed metrics penalise these systems for these errors and rank higher those that perform better on classes other than the majority one. For example, the winner according to *GMR*, *wF1* and *wF2* is NileTMRG that, according to Figure 2, shows higher accuracy on the *deny*, *support* and *query* classes, without considerably degraded performance on the majority class. *wAUC* still favours the Mama Edha system which has very limited performance on the important *deny* class. As is evident from Figure 2, NileTMRG is arguably the best system in predicting all classes: it has the highest accuracy for *denies*, and a sufficiently high accuracy for *support*, *queries* and *comments*. Using the same criteria, the second best system should be IITP. The only two metrics that reflect this ranking are *GMR* and *wF2*. In the case of *wF1*, the second system is UWaterloo,

⁴ $w_{support} = w_{deny} = 0.40$, $w_{query} = 0.15$ and $w_{comment} = 0.05$.

⁵The systems HLT(HITSZ), LECS, magc, UI-AI, shaheyu and NimbusTwoThousand are omitted because they do not provide the same number of inputs as the test set.

which has a very low accuracy on the *deny* class.

For RumourEval 2019, the best system according to macro- $F1$ (the official metric) is BLCU NLP, followed by BUT-FIT. However, after analysing the confusion matrices in Figure 3, we can conclude that eventAI is a more suitable model due to its high accuracy on *support* and *deny*. Metrics GMR , $wAUC$ and $wF2$ show eventAI as the best system. Finally, wshuyi is ranked as fourth according to GMR , $wAUC$ and $wF2$, while it ranked seventh in terms of macro- $F1$, behind systems like GWU and SINAI-DL that fail to classify all *deny* instances. Although wshuyi is clearly worse than eventAI, BLCU NLP and BUT-FIT, it is arguably more reliable than systems that misclassify the large majority of *denies*.⁶ Our analyses suggest that GMR and $wF2$ are the most reliable for evaluating RSC tasks.

4 Weight selection

In Section 3, $wAUC$, $wF1$ and $wF2$ have been obtained using empirically defined weights ($w_{support} = w_{deny} = 0.40$, $w_{query} = 0.15$ and $w_{comment} = 0.05$). These values reflect the key importance of the *support* and *deny* classes. Although *query* is less important than the first two, it is nevertheless more informative than *comment*.

Previous work tried to adjust the learning weights in order to minimise the effect of the imbalanced data. García Lozano et al. (2017) (Mama Edha), change the weights of their Convolutional Neural Network (CNN) architecture, giving higher importance to *support*, *deny* and *query* classes, to better reflect their class distribution.⁷ Ghanem et al. (2019) (UPV) also change the weights in their Logistic Regression model in accordance with the data distribution criterion.⁸ Nevertheless, these systems misclassify almost all *deny* instances.

Table 4 shows the RumourEval 2017 systems ranked according to $wF2$ using the Mama Edha and UPV weights. In these cases, $wF2$ benefits DFKI DKT, ranking it first, since *queries* receive a higher weight than *support*. However, this system only correctly classifies 6% of *support* instances, which makes it less suitable for our task than NileTMRG for instance. ECNU is also ranked

better than Mama Edha and IITP, likely due to its higher performance on *query* instances.

	$wF2$ Mama Edha	$wF2$ UPV
Turing	0.246 (8)	0.289 (8)
UWaterloo	0.283 (7)	0.322 (5)
ECNU	0.334 (3)	0.364 (3)
Mama Edha	0.312 (4)	0.349 (4)
NileTMRG	0.350 (2)	0.374 (2)
IKM	0.293 (5)	0.318 (7)
IITP	0.289 (6)	0.321 (6)
DFKI DKT	0.399 (1)	0.398 (1)

Table 4: RumourEval 2017 evaluated using $wF2$ with weights from Mama Edha and UPV.

Arguably, defining weights based purely on data distribution is not sufficient for RSC. Thus our empirically defined weights seem to be more suitable than those derived from data distribution alone, as the former accurately reflect that *support* and *deny* are the most important, albeit minority distributed classes. Further research is required in order to identify the most suitable weights for this task.

5 Discussion

This paper re-evaluated the systems that participated in the two editions of RumourEval task A (stance classification). We showed that the choice of evaluation metric for assessing the task has a significant impact on system rankings. The metrics proposed here are better suited to evaluating tasks with imbalanced data, since they do not favour the majority class. We also suggest variations of AUC and macro- $F\beta$ that give different weights for each class, which is desirable for scenarios where some classes are more important than others.

The main lesson from this paper is that evaluation is an important aspect of NLP tasks and it needs to be done accordingly, after a careful consideration of the problem and the data available. In particular, we recommend that future work on RSC uses GMR and/or $wF\beta$ (preferably $\beta = 2$) as evaluation metrics. Best practices on evaluation rely on several metrics that can assess different aspects of quality. Therefore, relying on several metrics is likely the best approach for RSC evaluation.

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⁶Confusion matrices for all systems of RumourEval 2019 are presented in Appendix A.

⁷ $w_{support} = 0.157$, $w_{deny} = 0.396$, $w_{query} = 0.399$ and $w_{comment} = 0.048$

⁸ $w_{support} = 0.2$, $w_{deny} = 0.35$, $w_{query} = 0.35$ and $w_{comment} = 0.1$

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A Confusion matrices for all RumourEval 2019 systems

For completeness, Figure 4 shows the confusion matrices of all systems submitted to RumourEval 2019. Apart from the four systems discussed in Section 3, all other systems fails to correctly classify the large majority of *deny* instances.

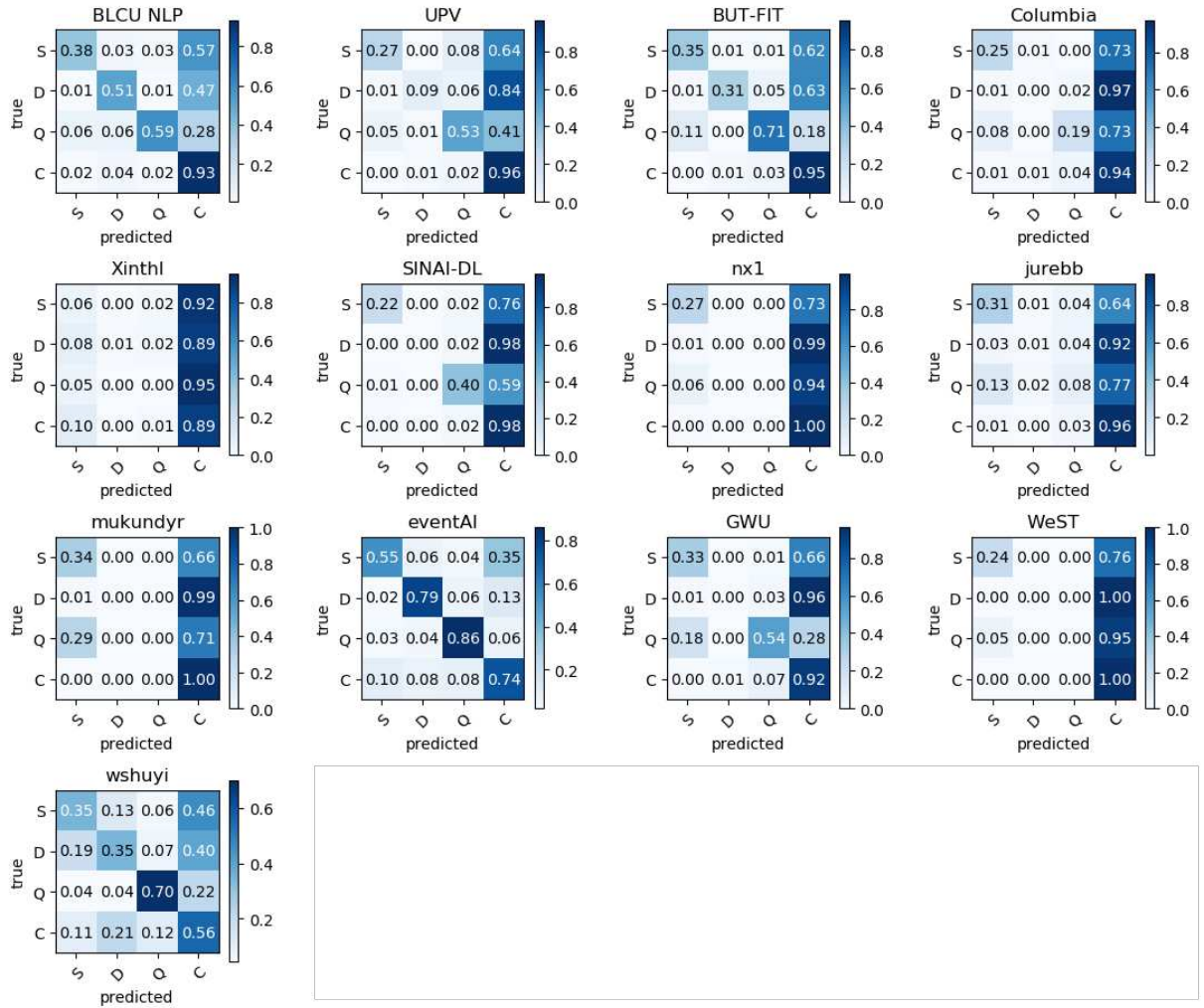


Figure 4: Confusion matrix for all systems from RumourEval 2019.